# Cognitive 2024 retake with solution

Kognitive Systeme (Technische Universität München)

Chair of Robotics, Artificial Intelligence and Real-Time Systems
Department of Computer Engineering
Technical University of Munich

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Registration number

Signature

**Note:**
- Cross your Registration number(with leading zero). It will be evaluated automatically.
- Sign in the corresponding signature field.



# Cognitive Systems

**Exam:** IN2222 / Retake **Date:** Monday 7th October, 2024
**Examiner:** Prof. Dr.-Ing. Knoll **Time:** 14:00 – 15:15

| | P 1 | P 2 | P 3 | P 4 | P 5 | P 6 |
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## Working instructions

- This exam consists of **20 pages** with a total of **6 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is **150 credits**.
- Detaching pages from the exam is prohibited.
- Allowed resources: None
- Write your name and matriculation number on the top of **every** page.
- **Give concise answers using appropriate terminology.**
- You may only answer in **English**.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- You have **75 minutes** working time

Left room from __________ to __________ / Early submission at __________

## Problem 1 Cognition and Cognitive Science (26 credits)

0 1 2 3

a) Briefly explain the difference between cognition and intelligence.

Cognition is a global process at the system level (1) that integrates (1) many different processing modalities.
Special cognitive skills (1), such as intelligence, learning, memory, interaction, etc., are constituents and synergies of a cognitive system.
*C02 1 Cognitive Systems Definition, slide 18*

0 1 2 3 4

b) Briefly state the two questions that are answered by the *proximate* and the *ultimate* explanation of a cognitive skill.

Proximate explanation: How is a certain behavior implemented in a cognitive system? (2)
Ultimate explanation: Why does a cognitive system exhibit a certain behavior? (2)
*C03 1 Basic Concepts in Cognitive Science, slide 19*

0 1 2 3

c) Explain the purpose, process, and success/failure evaluation of the Turing test.

- to determine if a computer can think (1)
- An interrogator communicates with a player through written notes, player is either a computer or a human. (1)
- The interrogator needs to tell the computer and human apart. (1)

*C02 1 Cognitive Systems Definition, slide 26*

0 1 2 3 4

d) List and explain two main properties of Dynamical Systems.

- Dissipation: the number of reachable states reduces over time
- Non-Equilibrium System: stable function requires an external "energy" supply"
- Non-Linearity: complex behavior can emerge from a small set of state parameters
- Collective Variables: the system is represented by a small set of state variables

1 point for every correct name and 1 point for every correct explanation
*C03 2 Basic Concepts in Cognitive Science, slide 38*

e) Name three approaches for Emergent Systems.

0 1 2 3

— Enactive Approach (1) — Connectionist Approach (1) — Dynamical Approach (1)
*C03 2 Basic Concepts in Cognitive Science, slide 24*

f) According to Vernon (2014), two of the cognitive capabilities required for a system to be considered self-reliant are Anticipation and Action.
Consider a self-driving car interacting with its environment. Choose any 3 of the *other* cognitive capabilities required by Vernon, define them, and describe how they apply in this case.

0 1 2 3 4 5 6 7 8 9

3 points each for any of (or similar to):

- Goal-Directedness (1):
  - take on goals, formulate predictive strategies to achieve them, and put those strategies into effect (1)
  - the car takes a destination as input and formulates a route (1)
- Autonomy (1):
  - operate with varying degrees of autonomy (1)
  - the car requires little to no input from the user (1)
- Interaction (1):
  - interact (cooperate, collaborate, communicate) with other agents (1)
  - communication with other self-driving cars to coordinate lane-changing and collision avoidance (1)
- Interpretation (1):
  - read the intention of other agents and anticipate their actions (1)
  - predict the short-term spatial behavior of other cars on the road (1)
- Sensing (1):
  - sense and interpret expected and unexpected events (1)
  - sensing nearby cars, road lines, and other obstacles (1)
- Reaction (1):
  - adapt to changing circumstances, in real-time, by adjusting current and anticipated actions (1)
  - reacting in real-time to obstacles, other cars losing control, adverse weather conditions (1)
- Learning (1):
  - from experience, adjust the way actions are selected and performed in the future (1)
  - over time learning traffic patterns to plan better routes (1)
- Anomaly detection (1):
  - notice when performance is degrading, identify the reason for the degradation, and take corrective action (1)
  - self-diagnosis of malfunctioning internal parts, reporting these problems to the user (1)

*C02 1 Cognitive Systems Definition, slides 22 23 24*

## Problem 2 Cognitive Architectures and Neurorobotics (17 credits)

0 1 2 3

a) Name three types of cognitive architectures.

Symbolic, Emergent, Hybrid

**E09 S12**

0 1 2

b) Explain the purpose of the MapSpikeSource and the MapSpikeSink devices of a Transfer Function in the Neurorobotics Platform.

- MapSpikeSink reads out spikes from the spiking neural network and converts them into a float value (1)
- MapSpikeSource creates spikes from a float value that is fed into the spiking neural network (1)

*Chapter E03, slides 27*

0 1 2 3 4 5 6

c) Describe the role of the Transfer Function (TF) in the Neurorobotics Platform (NRP).
Name four different tools that can be interactively used with the Transfer Function.

It connects and synchronizes brain models to robot models. (2)

- ROS/Gazebo
- OpenSim
- NEST
- Neuromorphic Hardware
- Custom Brain Models with Python Libraries such as Tensorflow, Pytorch
- Physical Robot in the real world

(one point per item, max four points)

**E03 S15-18**

d) In ACT-R, a production is a statement of a particular contingency that controls behavior. Every production is identified by a name and is assembled from a list of buffer operations.
The syntax of a production rule is illustrated below, where the parts before and after the "==>" symbol represent buffer tests and buffer actions, respectively.

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```
(p ProductionName
  buffer tests
==>
  buffer changes and requests
)
```

Based on the lecture, provide the two syntaxes used for

- Buffer Changes (Modifications),
- Buffer Requests.

```
=buffername> (1)
  ISA type (1)
  slot1 const1 (1)
  slot2 =var1

+buffername> (1)
  ISA type (1)
  slot1 const1 (1)
  slot2 =var1
```

(One point for each correct syntax provided for the buffer name, type, and any given slot)
**E09 S34-35**

## Problem 3 Human Brain (24 credits)

a) Explain whether it is true that only ten percent of the brain is used.

Not true; every part of the brain is used, but not all parts are used at once (2).
*Reference: C04-1 18*

b) Name the anatomical structure in the brain that plays a key role in forming episodic long-term memories.

Hippocampus (2)
**C04-2 S27**

c) Describe the purpose of glial cells.

Glial cells have a supporting function and do not play a direct role in information processing. (2)
**E04 S27**



d) In 1961, D. H. Hubel and T. N. Wiesel discovered two types of cells in the visual cortex, namely simple cells and complex cells.
Name the two important attributes of simple cells, i.e. name the attributes of features that do activate them.

- basic shapes, such as points or lines or edges
- correctly oriented
- correctly localized

One of the above solutions gives two points.
*Reference: C05-2 13*

e) Name and describe the two subsystems of the vertebrate nervous system.

Central Nervous System (CNS) (1) - It collects and distributes data throughout the body (1).
Peripheral Nervous System (PNS) (1)- The peripheral nervous system transmits signals between sensory organs, muscles and internal organs and the CNS (1).
**C04-1 S6**

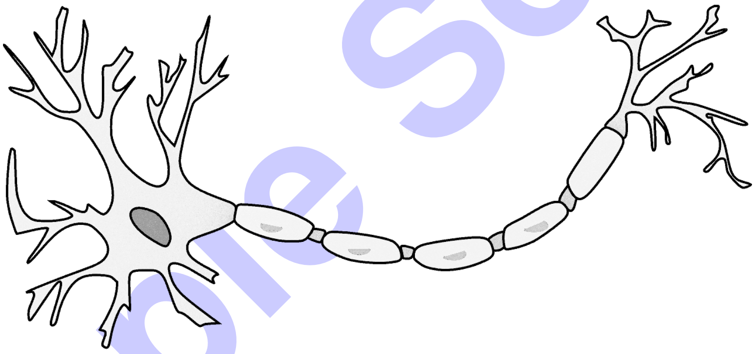
f) In 1943, Warren McCulloch and Walter Pitts proposed an artificial neuron model. Around ten years later Alan Hodgkin and Andrew Huxley described their biological neuron model.
Name two key differences between analog neuron models (artificial neuron models) and spiking neuron models (biological neuron models).

- Analog neuron models are functions that are not able to reproduce the temporal dynamics of biological neurons.
- Analog neurons do not generate spikes.
- dynamical system vs activation function
- digital spikes vs analog values
- rate vs real numbers
- high complexity vs low complexity
- high biological plausibility vs low

2 points per item, max 4 points
*(Reference: E04, Slide 38)*



g) Annotate the names of the three main components of a biological neuron in the sketch below.
Describe each component briefly.



- Soma (1): cell body containing the nucleus (1)
- Dendrites (1): receive electrical impulses (1)
- Axon (1): conducts electrical impulses (1)

*Reference: E04 S. 9*

## Problem 4 Biological Neurons and Spiking Neural Networks (30 credits)

0 1 2 3 4

a) In the lecture you have seen how the output of the general analog neuron can be calculated. Give the formula for the output of a general analog neuron and annotate the used variables.

The formula is given as (1)

$$n(\vec{x}) = A(\sum_i w_i x_i + b) = A(\vec{w}^T\vec{x} + b)$$

$\vec{w}$: weights (1)
$b$: bias (1)
$A()$: activation function (1)

**Reference:** EX03 - slide 35

0 1 2 3

b) Explain, why a multi-layer feed-forward network of general analog neurons **cannot** describe a non-linear function if the activation function is the identity, e.g. $A(z) = z$.

If we use the identity as the activation function, the formula simplifies to $n(\vec{x}) = \vec{w}^T\vec{x} + b$. If we build up a layer in a feed forward network, then the output of neuron $j$ is given as $n_j(\vec{x}) = \vec{w}_j^T\vec{x} + b_j$. Hence, we can compute the output of the entire layer as $\vec{n}(\vec{x}) = W\vec{x} + \vec{b}$, where $W$ is a matrix of dimension number of neurons $\times$ number of inputs (1). We see that the information is transformed by a simple matrix vector multiplication. It is a linear map. We know from linear algebra that the successive application of matrix vector multiplications is again a matrix vector multiplication. Hence, even with more layers, the network describes a linear transformation. (2)

**Reference:** EX03 - slide 35

0 1 2 3 4

c) State the two schemes of encoding techniques for spiking neural networks.
Name the main advantage of each of them.

- Rate coding (1): Robustness against noise (1) or
- Time coding (1): Fast reaction time (1) or energy efficient (1) or
- Population coding (1): robustness against noise (1) or

**EX05, slide 8**

The membrane potential of a simplified Leaky Integrate and Fire (LIF) neuron is described as a single differential equation

$$C\frac{dv}{dt} = I(t) - \frac{v(t)}{R}, \tag{4.1}$$

with membrane capacity $C$, membrane resistance $R$, and input current $I(t)$. Spikes of LIF neurons are generated if the membrane potential reaches a threshold value $v_{\text{thresh}}$. After the occurrence of a spike, the membrane potential is immediately set to $v_{\text{reset}} = v(0)$, i.e., we have no refractory period.

d) Name two essential features of neural processing that are captured by the *leaky integrate-and-fire* model.

0 1 2

natural permeability, passive ion channels, concentration gradient, membrane as capacitor

*Reference: Ex05 - problem sheet*

e) Solve the differential equation 4.1 for the case of a constant input current $I(t) = I$, given an initial membrane potential $v(0) = 0$ at initial time $t_0 = 0$.

0 1 2 3 4 5 6 7 8

- Separation of variables (2 points)

$$\int dv \frac{1}{v - RI} = \int dt \frac{-1}{R \cdot C}$$

- Correct integration (fraction to log) and limits (4 points)

$$\int_{v_0=0}^{v} dv \frac{1}{v - RI} = \int_{t_0=0}^{t} dt \frac{-1}{RC}$$

$$\Rightarrow \ln\left(\frac{v - RI}{-RI}\right) = -\frac{t}{RC}$$

- Correct result (2 points)

$$\Rightarrow v(t) = (-RI) \cdot \exp\left(-\frac{t}{RC}\right) + RI$$

*Reference: Ex05 - problem sheet*

0 1 2 3

f) The Hebbian theory can be summarized as "Cells that fire together wire together".
Name the three fundamental properties for a Hebbian learning algorithm to be biologically plausible and stable over an undetermined period of time.

Saturation (1), locality (1), or competition/selectivity (1)
*Reference: EX05 - slide 11*

0 1 2

g) In the context of Hebbian learning, describe what "saturation" means, and state the problem when a Hebbian learning algorithm does not possess this feature. You might draw a graph for illustrating the problem.

If a learning rule has saturation, the growth of the weights is limited to certain soft or hard boundaries. (Can also be explained with a graph) (2)

0 1 2 3 4

h) Equation (4.2) is a basic form of Hebbian learning.
Modify the equation (or add a second one) in order to add saturation to the learning rule.

$$w_t = w_{t-1} + \frac{\Delta t}{\tau_w} v \cdot u \tag{4.2}$$

A solution is to set a hard boundary that cannot be reached by the weights (2 points are only added if the equation is missing)

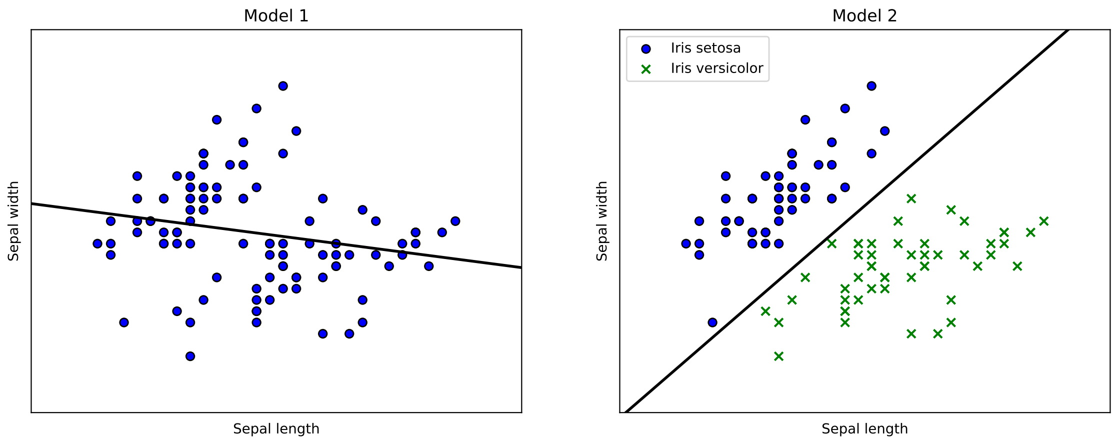
$$w_t = min\left\{w_{t-1} + \frac{\Delta t}{\tau_w} v \cdot u,\ w_{max}\right\}$$

(4 points)

*Reference: EX05 - slide 11*

## Problem 5 Machine Learning (23 credits)

You are assisting a colleague with machine learning projects using the Iris flower dataset. This dataset includes images, class labels, and features such as sepal length and width for 150 samples.
Below are two models that your colleague has developed from performing different tasks on the Iris dataset.




a) State the task performed by Model 1 and provide a brief explanation about this task.
Describe the inputs and outputs of this model.

0 1 2 3 4

- Model 1 is a regression model (**1p**).
- It means assigning a continuous value to a sample (**1p**).
- The input is the sepal length (**1p**), and the output is a continuous value for the predicted sepal width (or the other way around) (**1p**).

**C05-1: slide 11**

b) State the task performed by Model 2 and provide a brief explanation about this task.
Describe the inputs and outputs of this model.

0 1 2 3 4

- Model 2 is a classification model (**1p**).
- It means assigning a sample to one of the N discrete classes. (**1p**).
- The inputs are the sepal length and sepal width (**1p**), and the output is the predicted iris species (or class) (**1p**).

**C05-1: slide 10**

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c) Instead of using sepal width and length, your colleague opts to train a deep neural network to predict the species from raw flower images.

- State and explain two of the advantages deep neural networks offer over manually engineered features.

**4p** max.

- Deep neural networks can learn suitable feature space representations automatically (**2p**).
- Expert knowledge that is encoded in manually engineered features is replaced by huge datasets (**2p**).
- Layers in deep neural networks correspond to features at different levels of abstraction (**2p**).

**C05-2 p3**

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d) When designing the network, your colleague includes a max-pooling layer in the convolutional neural network.

- Explain what max-pooling is and one benefit it offers to the network.

Max-pooling preforms subsampling on the output of a convolutional layer by computing the maximum of a small subset of features (**2p**).
One of the following benefits (**2p**):

- Max-pooling reduces the size of the image patch and the number of synaptic weights in the following layers.
- Max-pooling makes the network more robust against translations in the input image.

**C05-2 p30**

e) Suggest one learning paradigm you will use to obtain Model 2.
Name and explain three other machine learning paradigms apart from this one.

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Model 1 uses supervised learning (**1p**).

- Unsupervised learning (**1p**): learning of statistical regularities in the input data (**1p**)
- Reinforcement learning (**1p**): learning which action to choose in a certain situation in order to maximize an external reward signal through interaction with the environment (**1p**)
- Semi-Supervised learning (**1p**): mixed dataset (labeled / unlabeled data) (**1p**)

**E01 p24**

## Problem 6 Reinforcement Learning (30 credits)

0 1 2 3 4 5 6

a) Describe classic conditioning and operant conditioning and state their main differences.

In classical conditioning, a subject learns the relationship between an initially neutral conditioned stimulus (CS) and an unconditioned stimulus (US) (1) that reflexively produces a conditioned response (CR) (1). When the subject is repeatedly presented the CS und US together, the CS will finally produce the CR even in the absence of the US. (1)

In operant conditioning, a subject learns the relationship between a stimulus and its behavior (1). A stimulus is only presented in response to a certain action of the subject (1) and serves as a reinforcer that increases or decreases the probability of that action. (1)

**C06 S 1:7**

0 1 2 3 4

b) The reward hypothesis determines the fundamental goal of reinforcement learning. Explain what the hypothesis is about.

Solution 1: That all of what we mean by goals and purposes can be well thought of as the maximization (1) of the expected value (1) of the cumulative sum (1) of a received scalar signal (called reward) (1).

Solution 2: RL's goal is to maximize (1) the expected value (1) of the total return $G_t = R_{t+1} + R_{t+2} + ...$ (2).

**C06 S 1:18**

c) Value functions are the central components in the value-based reinforcement learning methods taught in the lecture.

- Describe the value functions and their use in reinforcement learning.
- Name and define the two types of value functions and provide the mathematical equations of each.

0 1 2 3 4 5 6 7 8 9

A value function is a function that quantifies value for every state (1).
By accumulating the knowledge that the agent has gained from the experienced returns $G_t$ (1), the value function helps to find a policy $\pi$ that maximizes the return $G_t$ immediately at every time step $t$. (1)

1. State-value function (1): The expected return when a specific policy is followed after visiting a particular state (1).
$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s] \text{ (1)}$$
2. Action-value function (1): The expected return when a specific policy is followed after choosing an action in a particular state (1).
$$q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a] \text{ (1)}$$

**C06 S 2:4**

d) Briefly explain what is meant by bootstrapping in the context of reinforcement learning.
Furthermore, specify which of the following paradigms are based on bootstrapping.
→ Dynamic Programming / Monte Carlo / Temporal Difference

0 1 2 3

- Propagating value between consecutive states (1) by iteratively exploiting the recursive relationship that is formulated by the Bellman Equation (1) is denoted as bootstrapping.
**C06 S 3:4**
- Dynamic Programming bootstraps; Temporal Difference bootstraps; Monte Carlo doesn't bootstrap. (1)
**C06 S 6:6**

0 1 2 3

e) Briefly explain the problem with policies that do not explore.
Furthermore, explain how an $\epsilon$-greedy algorithm solves this problem.

- A non-exploratory policy may converge to sub-optimal solutions (1) as all state-action pairs are not guaranteed to be visited (1).
  **C06 S 4:5**
- **$\epsilon$-greedy**: Keeping a non-zero probability for every action (1) ensures an exhaustive exploration of all states to guarantee convergence toward an optimal policy.
  **C06 S 4:7**

0 1 2 3 4 5

f) Briefly explain the difference between on-policy and off-policy learning.
Furthermore, state

- the decisive advantage of off-policy learning,
- the necessary condition to have valid off-policy learning.

**On-policy**: The policy that is used to sample from the environment (behavior policy $b(a|s)$) and the policy that is used for the policy improvement step (target policy $\pi(a|s)$) are the same (1) (e.g., SARSA).

**Off-policy**: The policy that is used to sample from the environment (behavior policy $b(a|s)$) and the policy that is used for the policy improvement step (target policy $\pi(a|s)$) are different from each other (1) (e.g., Q Learning).

**Advantage of off-policy learning**: On-policy learning is actually a compromise as it learns action values not for the optimal policy, but for a near-optimal policy that still explores. By separating the optimal target policy from the exploratory behavior policy, off-policy learning helps in dealing with the exploration problem. (2)

**Condition of valid off-policy learning**: $\pi(a|s) > 0$ where $b(a|s) > 0$ (1)

**C06 S 5:6-7**

**Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.**